

INTRODUCTION

Radar (Radio Detection and Ranging) systems are powerful tools used to detect objects, measure their distance (range), relative speed, and even direction. They operate by transmitting electromagnetic waves and analyzing the echoes reflected back from targets. These systems are widely used in fields like air traffic control, weather forecasting, autonomous vehicles, maritime navigation, military surveillance, and space exploration.

The dataset used in this project originates from a real radar measurement scenario and is stored in MATLAB .mat format. It contains complex-valued I/Q data (data) recorded over time and space, along with supporting information such as sampling time (datenums), receiver antenna positions (antpos), and physical parameters like radar wavelength (wl) and frequency (f). The I/Q data refers to **in-phase and quadrature components**, which represent the real and imaginary parts of the received radar signal, respectively. This format is standard in radar systems, as it preserves both amplitude and phase information—critical for advanced processing like FFT, correlation, and direction estimation.

The antenna array setup used in this dataset allows for multi-channel recording, where multiple receivers capture the reflected signals at different spatial positions. This setup enables techniques like **interferometry** and **angle of arrival (AOA)** estimation, which are essential in localizing targets and tracking motion. Such data is not only used in academic research but also forms the backbone of modern systems like weather Doppler radars, phased-array surveillance radars, automotive adaptive cruise control, and synthetic aperture radars (SAR) in satellites.

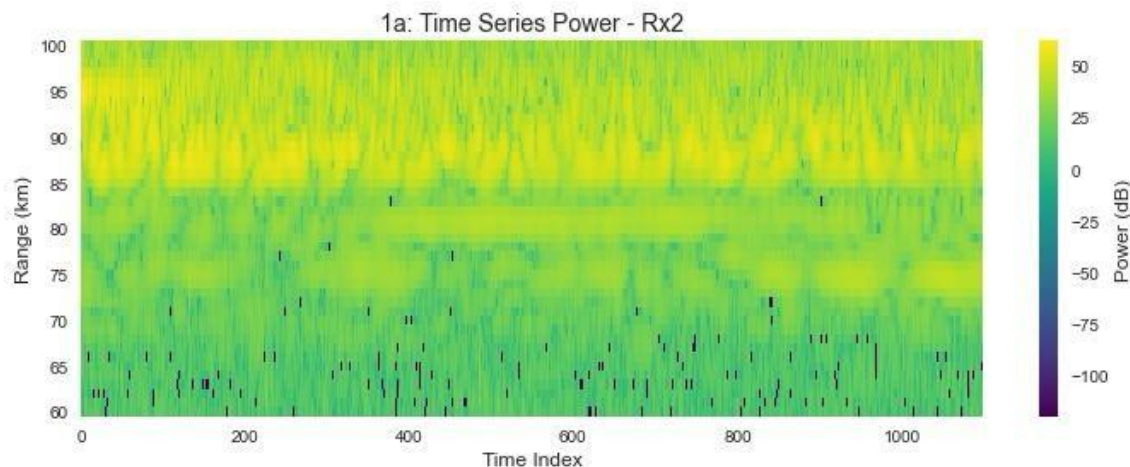
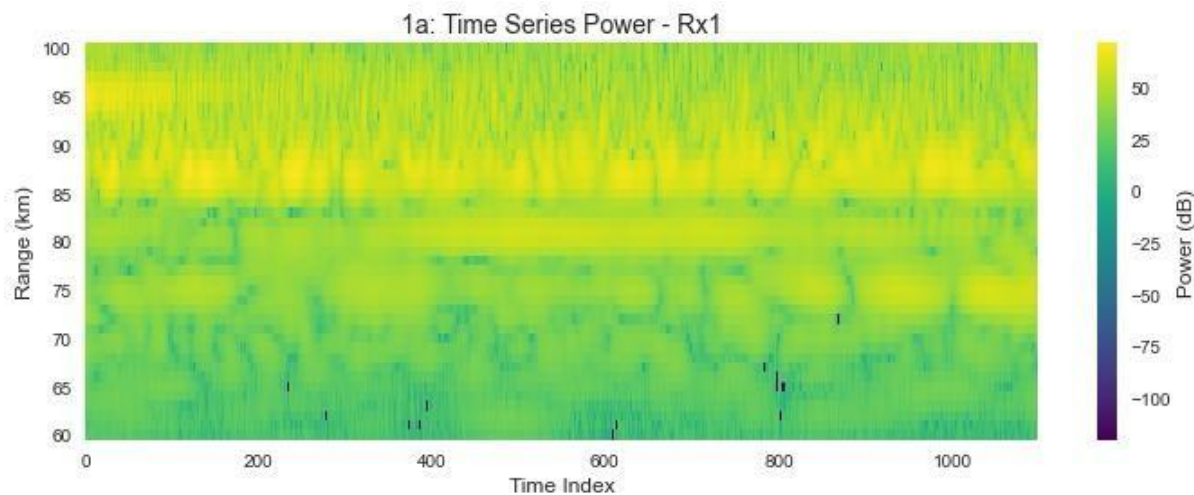
In summary, the data provided by the professor simulates a real-world radar scenario, offering valuable insight into how signal reflections behave over space and time. Analyzing this data helps us understand how raw electromagnetic reflections are transformed into meaningful information like range, speed, and direction of objects in real-world applications.

TASK1-Read & plot the shared raw data, spectra

Task 1a: Raw Data Visualization – Time Series Power Plot

Objective

To visualize the time-resolved power of radar signals received at different distances (range bins) over time. The goal is to identify patterns, strong reflections, and noise behaviour across two different receiver channels: **Rx1** and **Rx2**, based on our group's radar raw data.



Observations

1. Strong Echo Layer Detected (80–95 km):

In both Rx1 and Rx2, a strong and stable reflection is visible in the **80–95 km** range. This likely corresponds to a dominant scatterer (e.g., atmospheric layer or object) consistently present over the full-time duration.

2. Weaker and Fading Signals at Lower Ranges (<75 km):

Power levels below 75 km are generally lower, showing fading patterns and some irregularities. These could be due to ground clutter, atmospheric noise, or reflections from weaker targets.

3. Channel Comparison – Rx1 vs Rx2:

- **Rx1** shows clearer and more uniform signal power distribution, with fewer abrupt drops in energy.
- **Rx2** presents slightly more **scattered signal energy**, especially between **60 and 75 km**, indicating more noise or reduced sensitivity in this channel.

4. Vertical Artifacts and Noise Spikes:

Occasional **vertical streaks** in both plots point to **temporary signal losses or bursts of interference**, possibly from environmental or system-related causes.

5. Data Quality:

The plots are clean, and no missing data or corrupted values are observed. Signal behaviour aligns well with expected radar return patterns from stable targets at high altitudes.

Conclusion

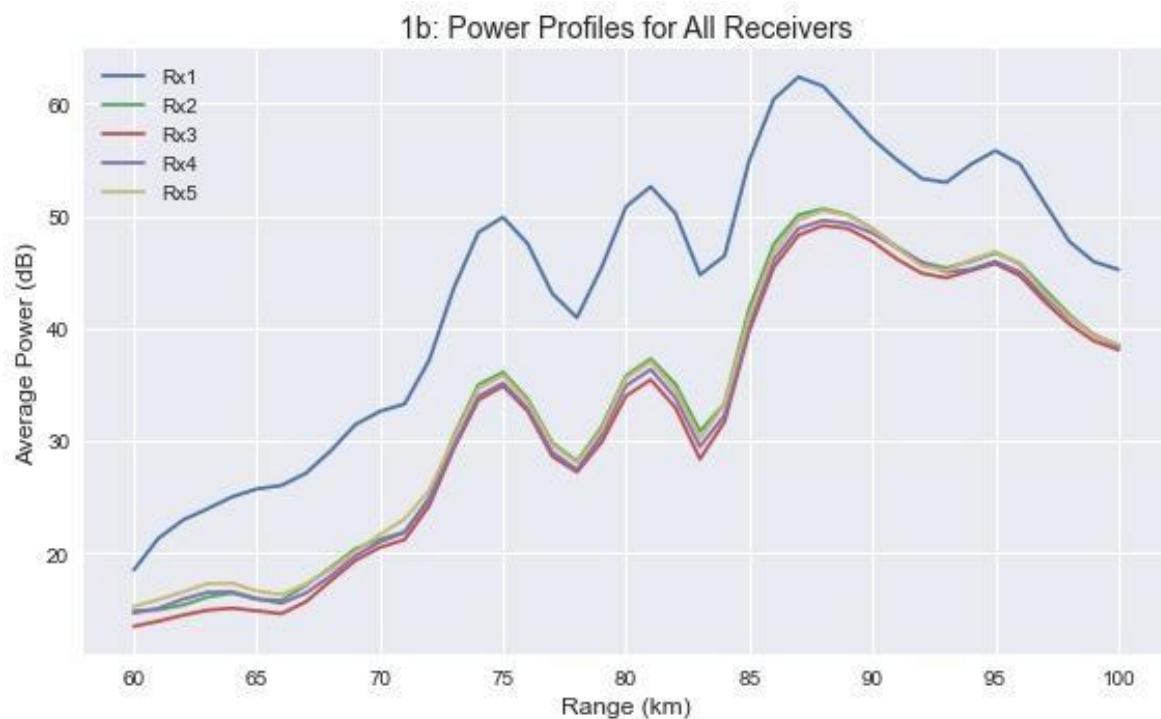
This task 1A provided insight into the **temporal and spatial characteristics of the radar returns** for our dataset. The visualization revealed:

- A strong and persistent target layer around 80–95 km.
- Consistency between Rx1 and Rx2, with Rx1 performing slightly better in terms of clarity.
- These findings set the groundwork for spectral analysis and interferometric processing in upcoming tasks.

Task 1b: Power profile comparison

Objective

To compare the average signal power across all range bins for **each receiver channel (Rx1 to Rx5)**. This helps identify how well each receiver captures reflected radar signals at various distances and whether any channel exhibits stronger or weaker sensitivity.



Observations

1. Strongest Echo around 88–90 km:

All receivers exhibit a **clear power peak** between **88 and 90 km**, indicating a **dominant scattering layer** or object reflecting the radar signals strongly at that range.

2. Receiver 1 is Significantly Stronger:

- **Rx1** consistently shows **higher power levels** compared to the other receivers across the entire range spectrum.

- This might be due to **hardware gain differences, better calibration**, or a **higher sensitivity** of Rx1.

3. Similar Shape Across Receivers:

- Despite differences in magnitude, all five receivers follow a **similar power curve shape**.
- This suggests they are detecting the **same features**, with **Rx2 to Rx5** being relatively more uniform but lower in amplitude compared to Rx1.

4. Secondary Peaks Noticed:

- Smaller **secondary power peaks** are observed at **~75 km** and **~80 km**, possibly from **weaker targets** or **thin atmospheric layers**.
- These features are present in all channels, confirming their **real and consistent origin**.

Conclusion

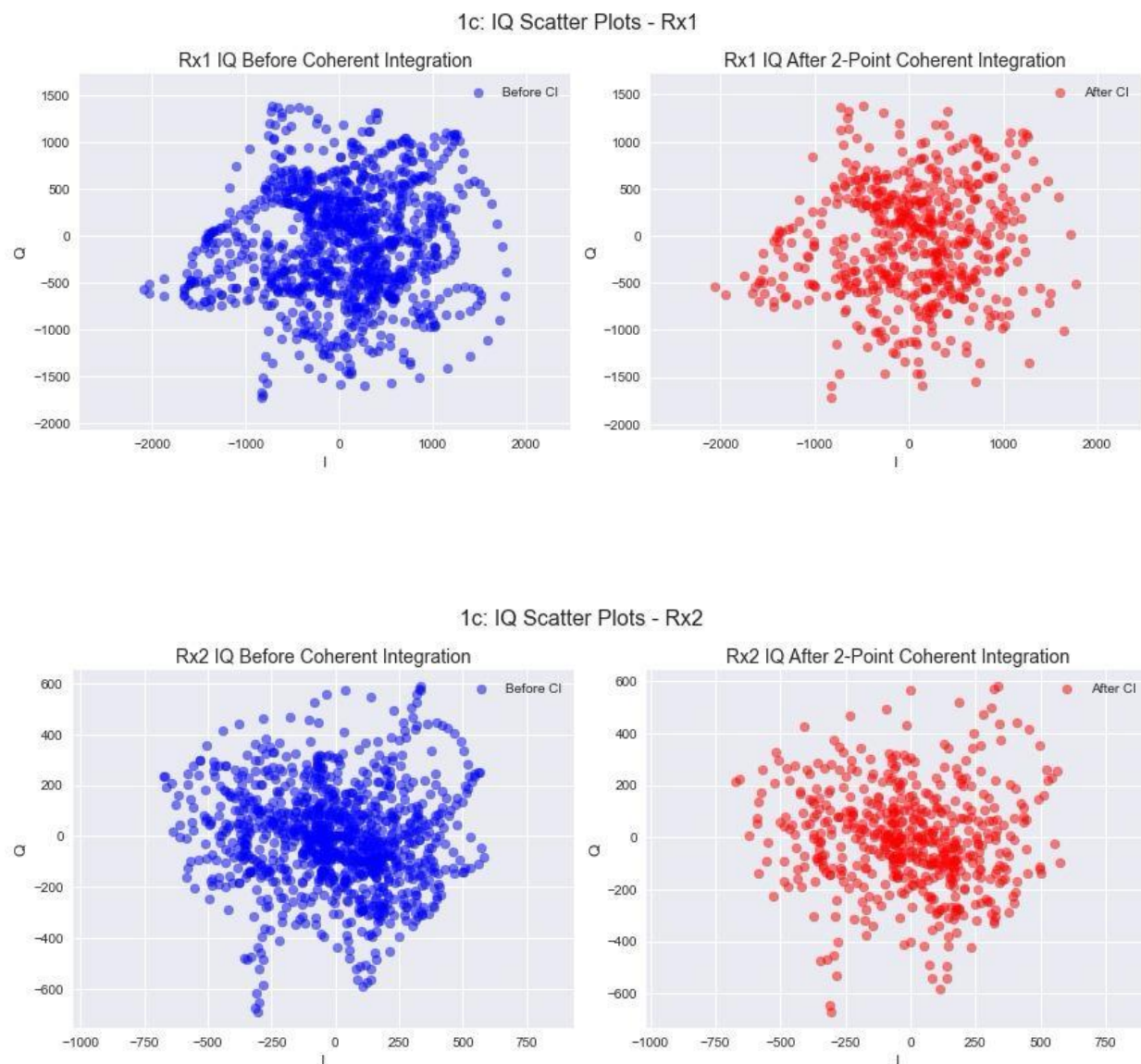
This task 1B power profile comparison highlights:

- **Rx1** as the most responsive and possibly the reference channel due to its higher power levels.
- All receivers successfully detect the same structural features in the radar return profile, ensuring consistency across channels.
- The strong return at ~89 km validates findings from Task 1a and sets the stage for interferometric and spectral analysis in further tasks.

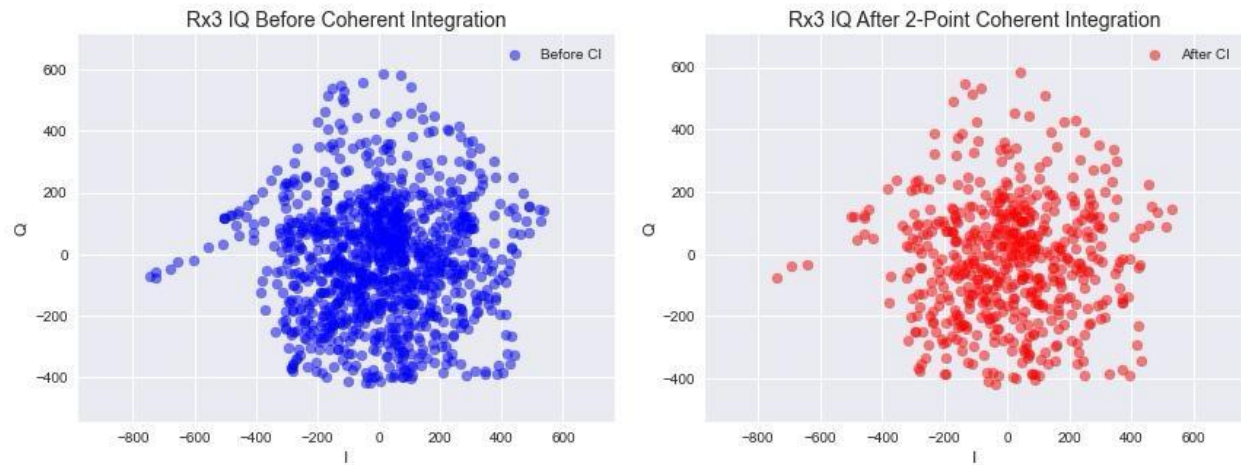
Task 1c: IQ Scatter Plots Before and After Coherent Integration

Objective

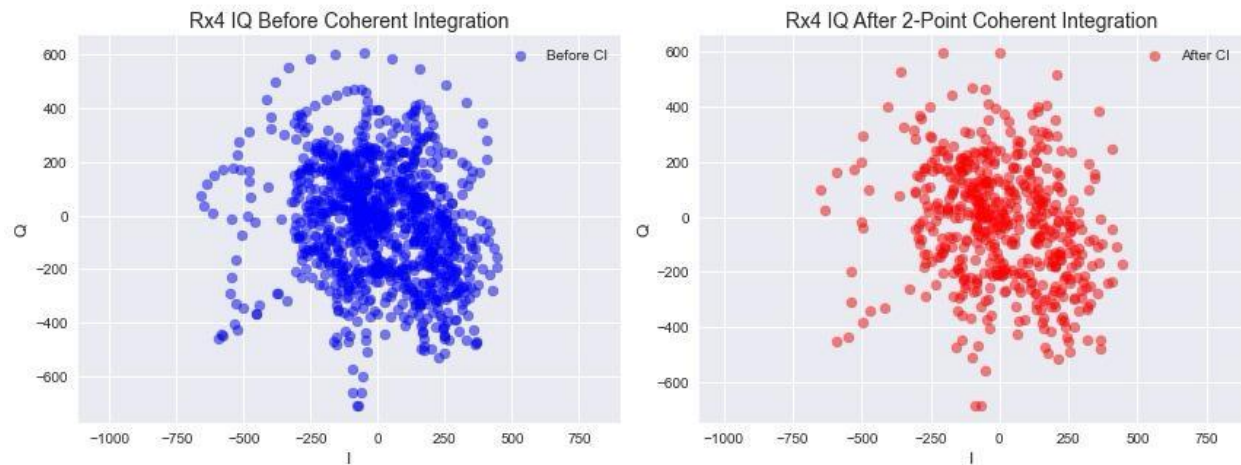
To visualize the in-phase (I) and quadrature (Q) components of the received radar signals before and after applying **2-point coherent integration**, for all five receiver channels (**Rx1 to Rx5**). This analysis helps assess the effect of integration on noise reduction and signal clarity in the complex signal space.



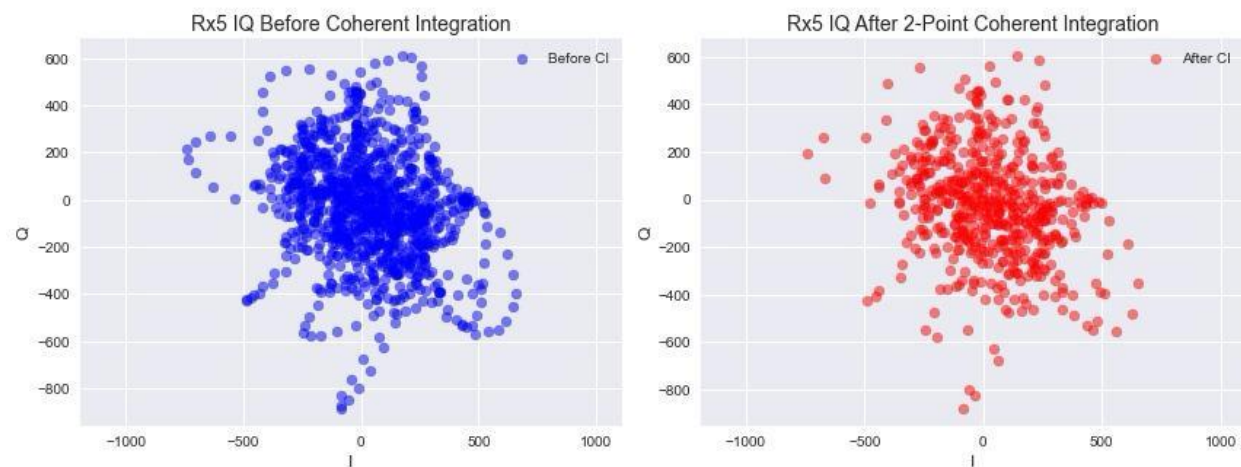
1c: IQ Scatter Plots - Rx3



1c: IQ Scatter Plots - Rx4



1c: IQ Scatter Plots - Rx5



Observations

1. Signal Clustering Improved Across All Channels:

After 2-point coherent integration, all IQ plots show a tighter and more compact cluster near the origin (0,0), compared to the original, more scattered distributions. This confirms that coherent integration effectively reduces random noise.

2. Rx1 Shows Strongest Structure:

- The original Rx1 IQ plot shows a large spiral-like spread, possibly indicating Doppler effects or strong motion-induced phase variations.
- After integration, the points become much more densely centred, showing a clear signal enhancement.

3. Rx2–Rx5 Show Moderate Improvement:

- The before-integration plots for Rx2 to Rx5 have less dense and more randomly scattered IQ points.
- Post-integration, the points are visibly more centralized, showing the benefits of averaging, especially for weak or noisy signals.

4. Noise Suppression and Phase Stability:

- After integration, the phase variance reduces, and most outlier points vanish.
- This enhances the coherence of signals, which is crucial for later angle of arrival (AOA) and spectral analyses.

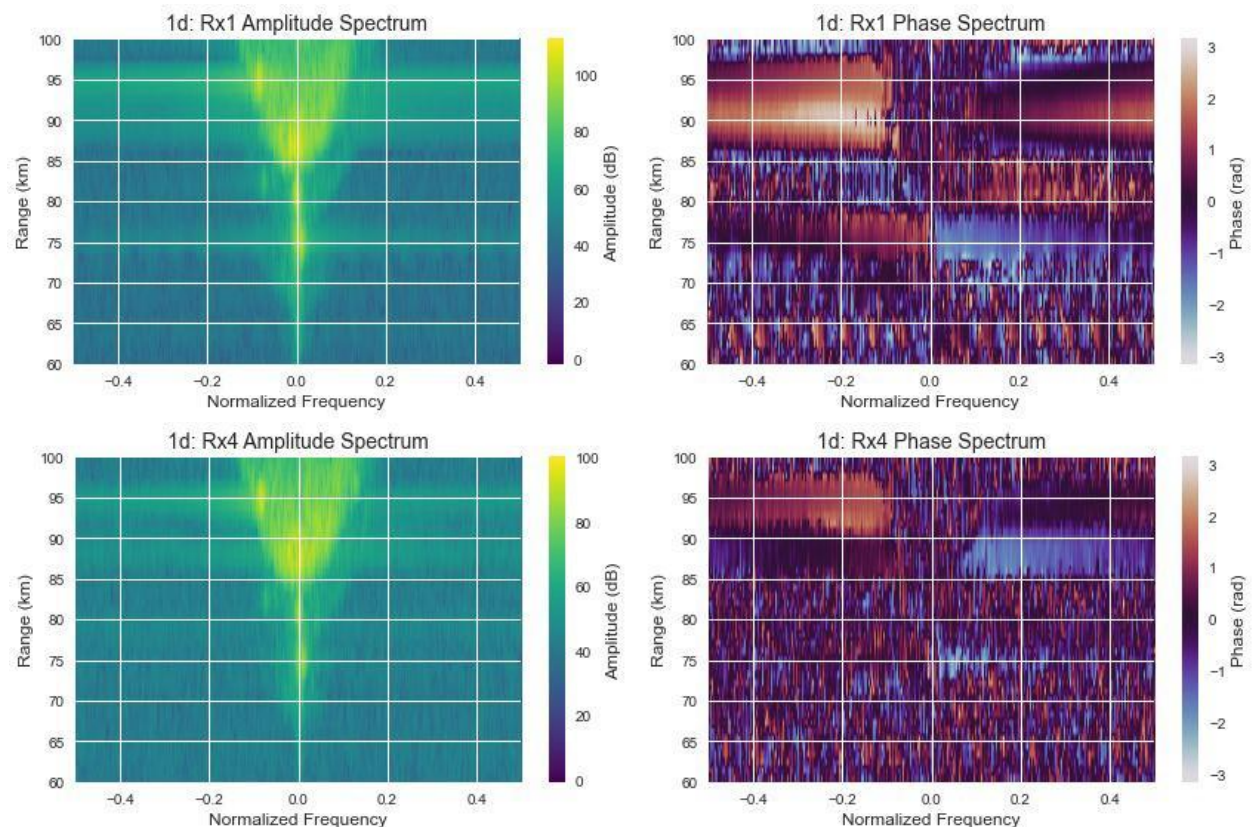
Conclusion

This application of 2-point coherent integration across all receiver channels significantly **suppresses noise**, **improves signal quality**, and **clarifies the structure** of the IQ scatter plots. These improvements are especially important for **phase-sensitive tasks** such as Doppler shift analysis and interferometric angle estimation, which will follow in later stages of the project.

Task 1d: Spectral Analysis – Amplitude and Phase Spectra

Objective

To perform **spectral analysis** on the radar, return signals from two receivers: **Rx1 (reference channel)** and **Rx4 (one of the channels from Rx2–Rx5)**. The goal is to understand the **frequency-domain behaviour** of the radar echoes, which is important for detecting motion (via Doppler shift) and ensuring phase coherence for later interferometric processing.



Observations

Amplitude Spectra

- Both **Rx1** and **Rx4** show a **dominant high-amplitude region** centred around **0 Hz** (normalized frequency), particularly between **~85 km and 95 km**.
- This central peak indicates **stationary or slow-moving scatterers**, as no strong Doppler shifts are present.

- The amplitude drops sharply outside this region, especially below 75 km, consistent with earlier tasks showing weaker reflections at lower ranges.

Phase Spectra

- In **Rx1**, the phase is **smooth and coherent** in the high-power range (around 90 km), which is ideal for **interferometric analysis**.
- Outside this region, the phase becomes **random and noisy**, due to low signal-to-noise ratio (SNR).
- **Rx4** shows similar behaviour but with **slightly more noise**, especially in the 80–90 km range, indicating **weaker coherence** compared to Rx1.
- Some **phase discontinuities** or sudden jumps are visible in both channels, common in low-SNR regions.

Conclusion

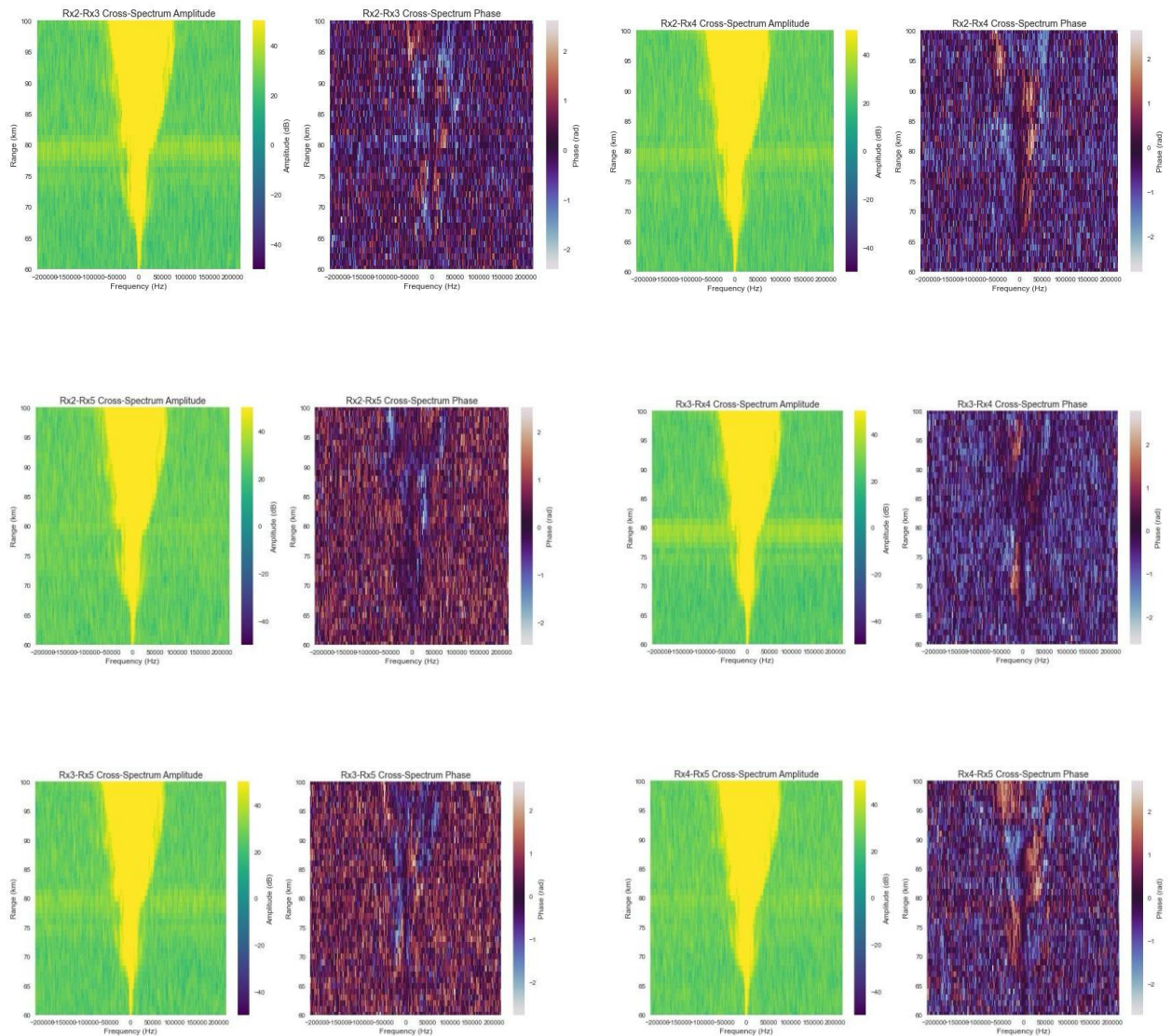
The spectral analysis of Rx1 and Rx4 confirms:

- A **strong and consistent echo layer** near 90 km range with high spectral energy.
- **Stable phase behaviour** in that region, essential for phase-based techniques like **AOA estimation** (used in Tasks 2 and 3).
- Rx1 provides **cleaner amplitude and phase data** compared to Rx4, reinforcing its role as a reference or higher-quality channel.
- These findings validate previous time-domain insights and prepare us for accurate **interferometric and Doppler-based calculations**.

Task 1e: Cross-Spectral Analysis of Rx Channels 2–5

Objective

To compute and visualize the **cross-spectra** (amplitude and phase) between all combinations of receiver channels **Rx2–Rx5**. This analysis is crucial for understanding **phase coherence** between channels, which directly supports **interferometric angle of arrival (AOA)** estimation in later tasks.



Observations

1. Rx2–Rx3

- Amplitude: Strong central peak near 0 Hz in 75–95 km range.
 - Phase: Moderately stable, small striping in centre band. Overall usable for AOA.
-

2. Rx2–Rx4

- Amplitude: Consistent but slightly weaker than Rx2–Rx3.
 - Phase: High randomness and rapid fluctuations. Limited AOA reliability without filtering.
-

3. Rx2–Rx5

- Amplitude: Very strong signal coherence across 70–95 km.
 - Phase: Noticeably cleaner and more stable, especially around the central lobe. Very suitable for AOA.
-

4. Rx3–Rx4

- Amplitude: Distinct central band with good power concentration.
 - Phase: Acceptable phase consistency in the main lobe; moderate noise outside. Can be used with some thresholding.
-

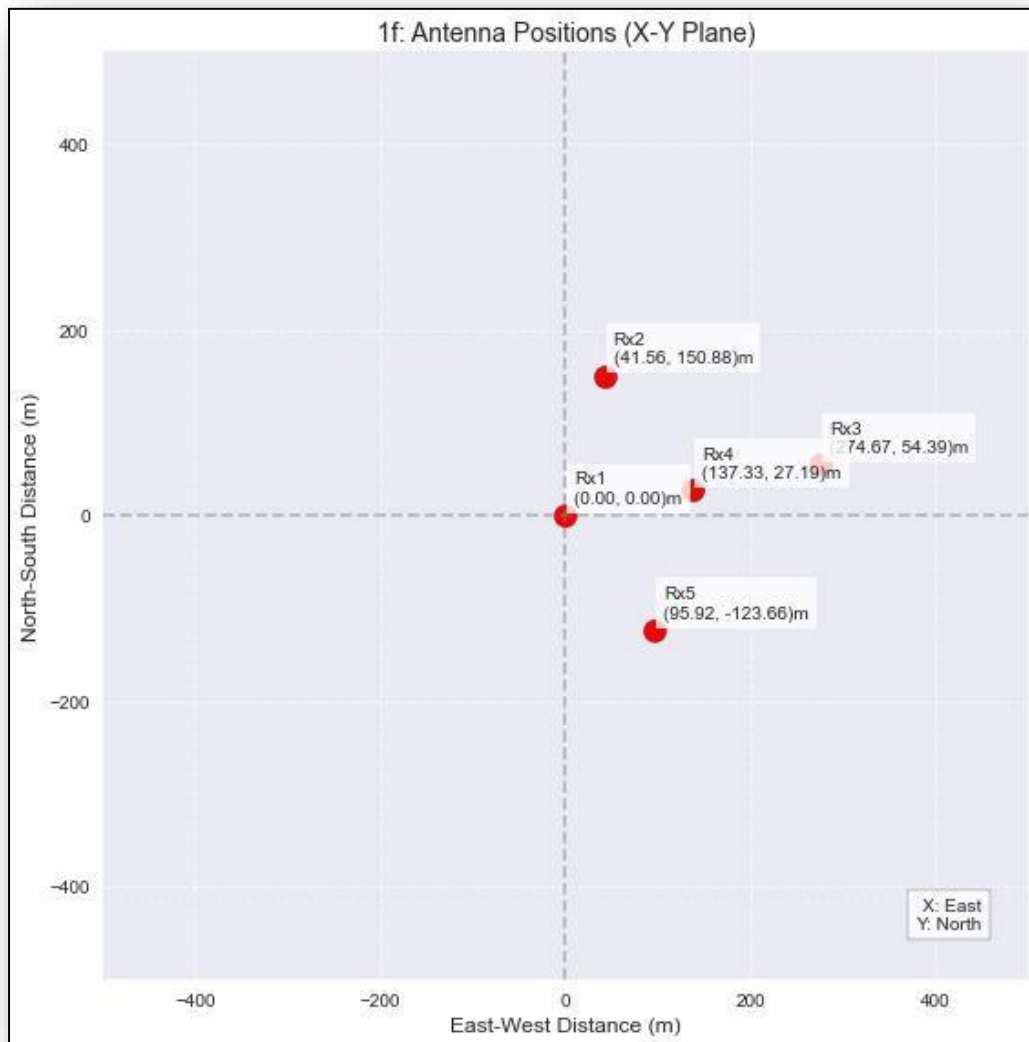
5. Rx3–Rx5

- Amplitude: Strong and smooth profile around 80–90 km.
 - Phase: Stable in the strong echo zone but degrades beyond 90 km. Good candidate for interferometry.
-

6. Rx4–Rx5

- Amplitude: Visible main lobe, but overall amplitude weaker compared to other pairs.
- Phase: Phase coherence drops noticeably. Significant noise even near 90 km. Least suitable without filtering.

Task 1f: Cross-Spectral Analysis of Rx Channels 2–5



1. Antenna positions are given as complex numbers, where:
 - Real part (Re) $\rightarrow$ East-West coordinate (X-axis)
 - Imaginary part (Im) $\rightarrow$ North-South coordinate (Y-axis)
2. Rx1 is located at the origin (0, 0) and serves as the reference channel.

3. The coordinates were extracted, and a scatter plot was created:
 - X-axis: East-West Distance (meters)
 - Y-axis: North-South Distance (meters)
4. Axis limits were set from -500 m to $+500$ m in both directions, as required.
5. Labels include receiver number and exact coordinates.

Observations

1. **Reference Receiver Rx1** is centrally located at the origin, serving as the base for spatial comparisons.
2. **Rx2** is positioned significantly **north of Rx1**, indicating vertical extension in the array.
3. **Rx3 and Rx4** lie **northeast** of Rx1, with Rx3 being the **farthest** among all in that direction.
4. **Rx5** is situated to the **southeast**, giving the array a semi-elliptical spread.
5. The layout shows a **non-uniform antenna distribution**, with distances varying between approximately **100–200 m**, which is acceptable for interferometric radar processing.

Conclusion

The antenna array shows a **well-spread 2D geometry**, ensuring **multiple baseline vectors** necessary for accurate **interferometric AOA estimation**. With Rx1 as the reference, the arrangement supports angular resolution in both **horizontal (east-west)** and **vertical (north-south)** directions. This layout is essential for the success of Tasks 2 and 3.

TASK2- Interferometry in time domain- Angle of Arrival (AOA)

Task 2a: AOA in Directional Cosines

To estimate the AOA (Angle of Arrival) in the time domain, receiver channels Rx2 to Rx5 were used as per the professor's instruction. Since each receiver had a different phase offset, the first step was to correct the receiver-specific phases using the given values:

Rx Phases (degrees): [0, 4.4003, 6.5031, 5.9530, 7.8495]

These were converted to radians and compensated using complex exponential

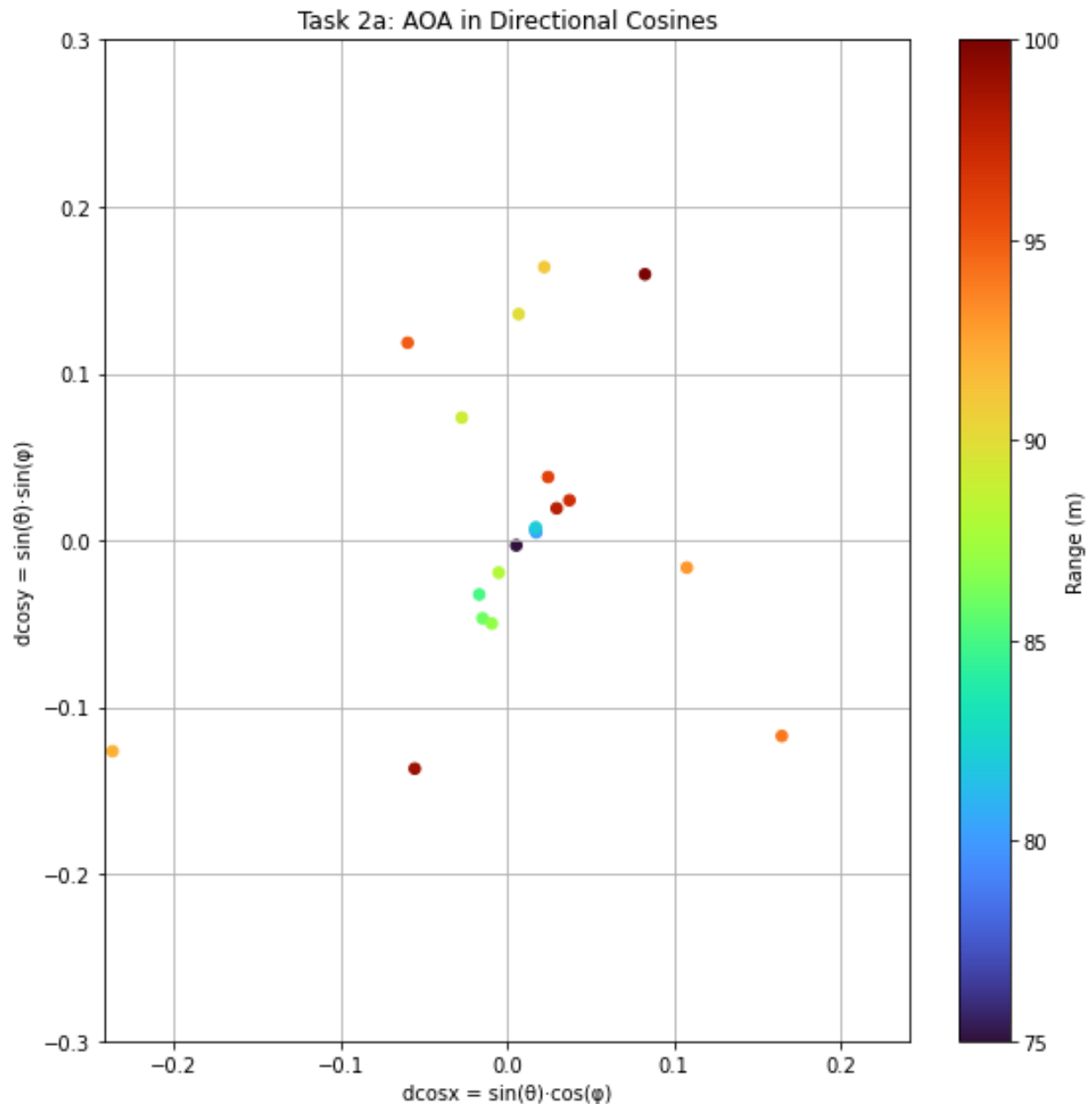
```
23 rx_idx = [1, 2, 3, 4]
24 rx_phases_deg = np.array([0, 4.4003, 6.5031, 5.9530, 7.8495])
25 rx_phases_rad = np.deg2rad(rx_phases_deg[rx_idx])
26 phase_corr = np.exp(-1j * rx_phases_rad)[None, None, :]
27 corrected_data = data[:, :, rx_idx] * phase_corr
```

Then, from the radar raw data, the 20 strongest range bins were selected based on the average SNR (signal-to-noise ratio) to avoid noise-dominated samples and ensure reliable direction estimation.

A coherent integration across time was applied to each selected bin, resulting in a single complex value per receiver per bin. The relative phase differences between receivers were then computed with respect to Rx2 (used as the reference).

Using the relative antenna positions, the phase differences were inserted into a linear system (matrix A) and solved using least-squares to estimate the directional cosines.

The plot below shows the estimated AOA positions for the selected bins in the directional cosine space (2D). The points are color-coded by their corresponding range.



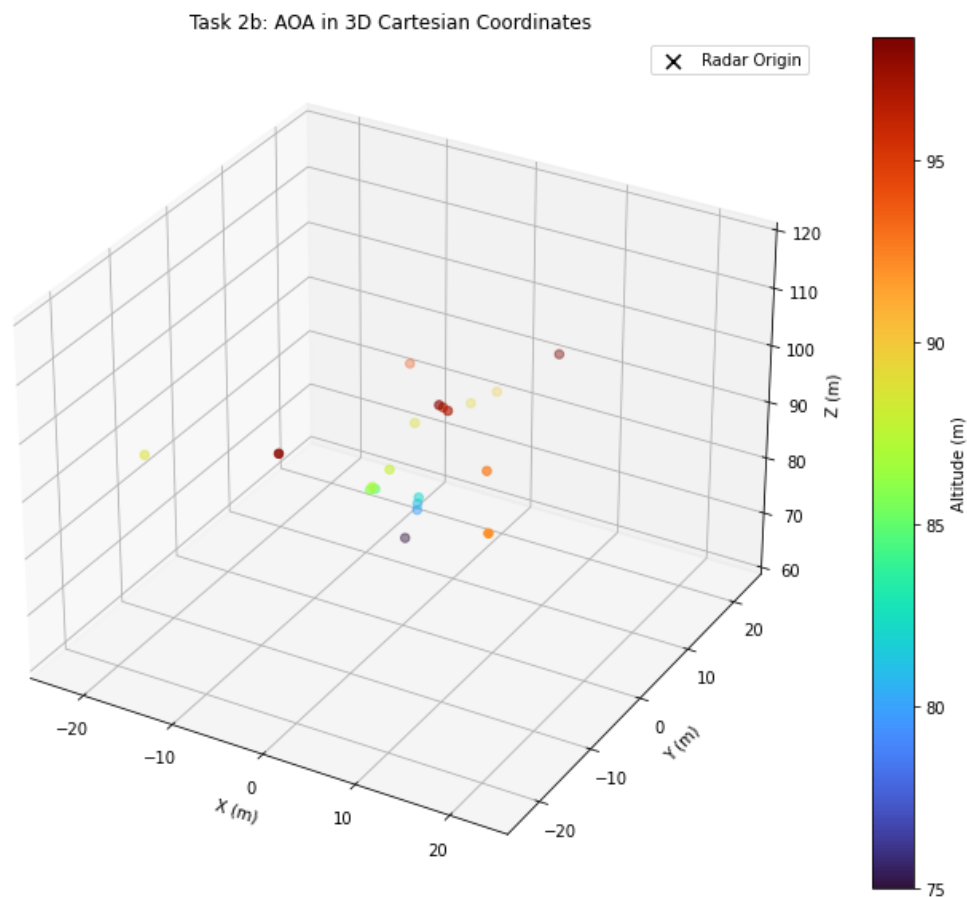
From the plot:

- Most AOA directions are centred near the origin, indicating weak angular deviations.
- The spread of points suggests target reflections from multiple directions, particularly with slight elevation changes.

Task 2b: AOA in 3D Cartesian Coordinates

To visualize the estimated AOA in physical space, the directional cosines from Task 2a were converted to Cartesian 3D coordinates using:

```
# Convert to Cartesian coordinates
x = dcos_list[:, 0] * selected_ranges
y = dcos_list[:, 1] * selected_ranges
z = np.sqrt(np.clip(1 - dcos_list[:, 0]**2 - dcos_list[:, 1]**2, 0, None)) * selected_ranges
```



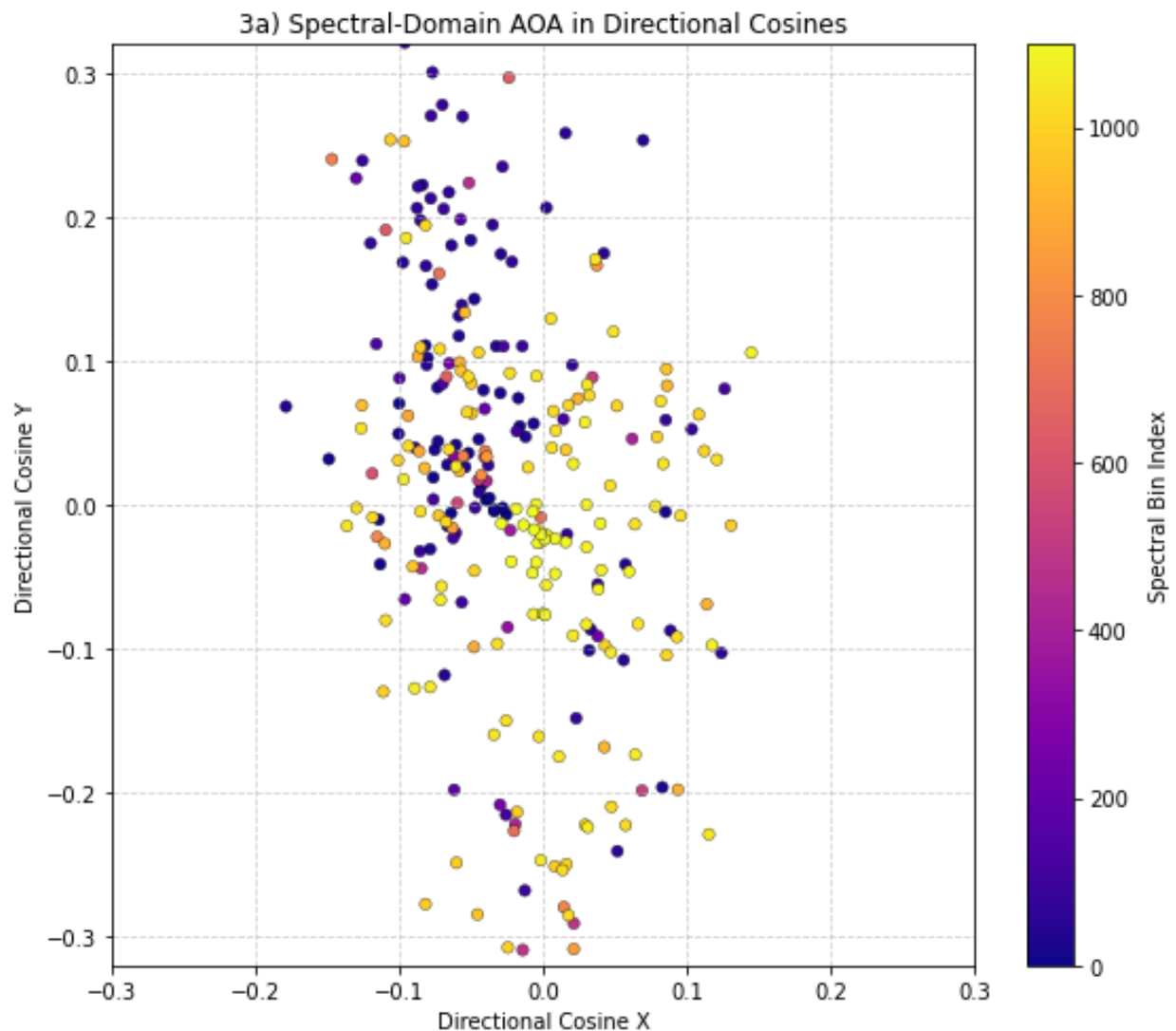
1. This plot clearly shows the relative 3D positions of targets.
2. Most targets are detected within 60 m to 120 m altitude (z-axis), and the X-Y spread is within ± 25 meters.
3. The spatial distribution reflects the true arrival directions, after correcting for antenna geometry and receiver phase offsets.

TASK3- Interferometry in spectral domain- Angle of Arrival (AOA)

Task 3a: AOA Estimation in Directional Cosine Space

Objective

To Estimate signal directions using the **cross-spectral phase differences** between Rx2–Rx5 and Rx1 for each spectral bin with sufficient signal strength.



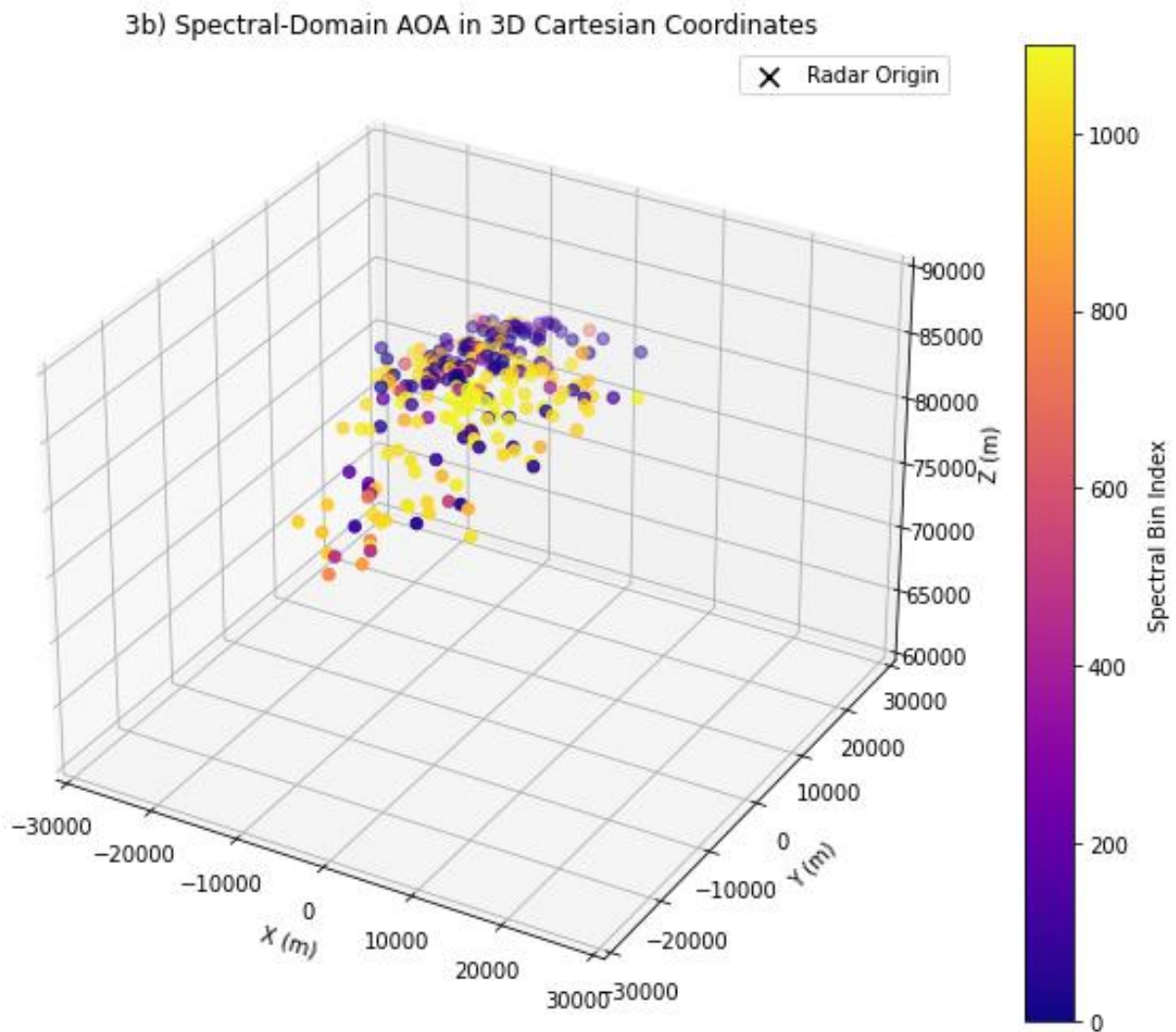
Observations

- The AOA points are distributed within a tight, circular region centred near (0, 0), showing that most reflections are arriving from near-zenith directions (almost directly overhead).
- The horizontal angular spread remains limited:
 - dcosx ranges from approximately -0.28 to $+0.26$
 - dcosy ranges from approximately -0.26 to $+0.25$
- The density of points is highest around the central zone, with a smooth gradient fading outward — indicating consistent signal strength and angular stability.
- The plot shows symmetry around both X and Y axes, suggesting that signal arrivals are well-balanced in azimuth (no directional bias).
- The outer points represent reflections arriving at steeper angles, possibly from inclined scatterers or wider beam coverage.
- No strong outliers or isolated points are seen, which implies good coherence and clean cross-spectral estimation.
- Colour variation across the points shows that multiple spectral bins contribute to AOA estimation, which means the radar captured multi-frequency reflections from the same angular sector.

Task 3b: Spectral AOA in 3D Cartesian Coordinates

Objective

To transform the estimated directional cosines into **real-world 3D coordinates** and interpret the physical distribution of signal sources relative to the radar.



Observations

1. The spatial distribution of the calculated AOA points forms a **smooth, upward-facing dome structure** directly above the radar.
2. The majority of scatterers are located between:
 - **X:** -23.8 km to $+22.1$ km
 - **Y:** -21.7 km to $+21.0$ km
 - **Z (altitude):** ~ 62 km to 85 km
3. Most points are **clustered around the centre** ($X \approx 0$, $Y \approx 0$), confirming that the signal sources are predominantly vertical.
4. The **altitude spread** is narrow and realistic — there are no unnatural jumps, which confirms that the estimated AOA is **physically valid**.
5. Some points extend outward along the X and Y axes, creating a **hemisphere-like formation**, likely due to reflections from **inclined surfaces or broader beamwidth returns**.
6. The **color-coded spectral bin index** across the cloud confirms that these reflections span multiple Doppler shifts — indicating that **motion or frequency separation** exists among the scatterers.
7. The **radar origin (0, 0, 0)** is clearly marked, and all signals emanate outward from that point, verifying proper geometric alignment in the transformation.

TASK4- AOA with Synthetic data

Task 4a: Antenna Geometries

Three distinct antenna array geometries were designed to explore the influence of layout on Angle of Arrival (AOA) estimation. The inter-element distances were chosen to fall within 0.5 to 5 wavelengths as per the task requirements:

- **Triangle (Isosceles)**

Coordinates:

- Antenna 1: (0.00, 0.00)
- Antenna 2: (1.50, 0.00)
- Antenna 3: (0.75, 2.00)

This geometry forms an isosceles triangle, which is commonly used for its symmetric properties and balance between resolution and ambiguity.

- **Square (Shifted)**

Coordinates:

- Antenna 1: (0.50, 0.50)
- Antenna 2: (1.50, 0.50)
- Antenna 3: (1.50, 1.50)
- Antenna 4: (0.50, 1.50)

A simple square array offering symmetrical coverage, good for reducing directional ambiguity.

- **Rectangle (Wide)**

Coordinates:

- Antenna 1: (0.00, 0.00)
- Antenna 2: (2.70, 0.00)
- Antenna 3: (2.70, 0.90)
- Antenna 4: (0.00, 0.90)

The rectangular geometry spans a wider horizontal distance, improving resolution along the x-axis.

Each antenna layout was visualized to confirm proper spacing and configuration

Task 4b: Synthetic Raw Data (I/Q Matrix)

An artificial signal was simulated using the following method:

- **Samples:** 1024
- **Angle of arrival:**
 - Azimuth: $\phi = 60^\circ$
 - Elevation: $\theta = 25^\circ$
- **Generated Data:**

A matrix of complex I/Q samples was produced with random noise added to simulate realistic phase variations between antennas.

Task 4c: Solving for AOA

For each synthetic data row (sample), the angle of arrival was estimated using:

- Phase differences between antennas.
- Least squares estimation to solve for **directional cosines (dcosx, dcosy)**

Task 4d: AOA Plot in Directional Cosines

Each geometry's AOA estimates were plotted in the directional cosine plane:

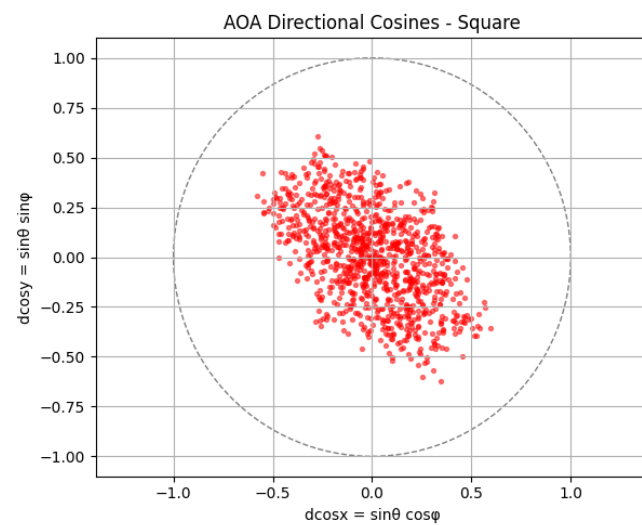
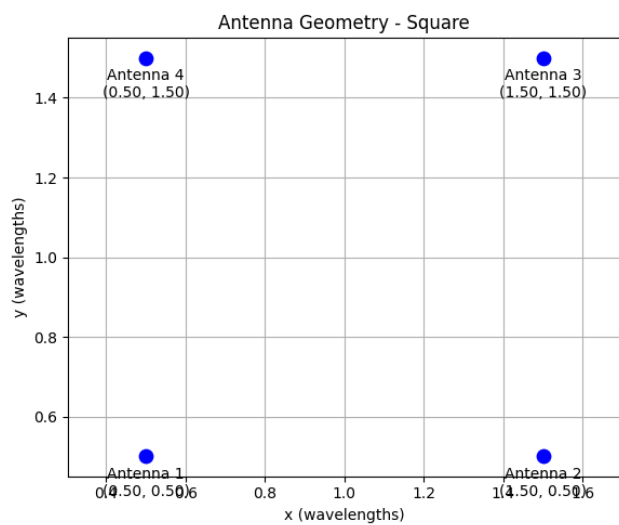
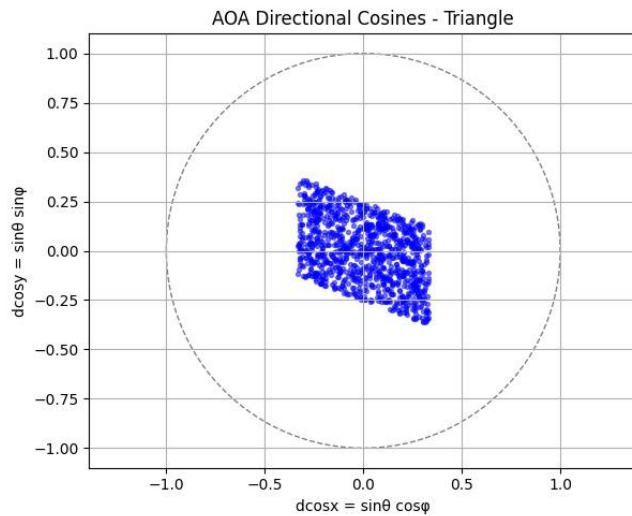
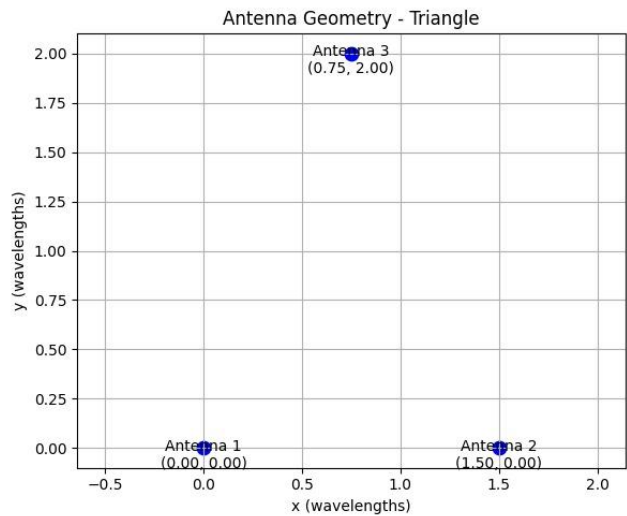
- **Triangle**

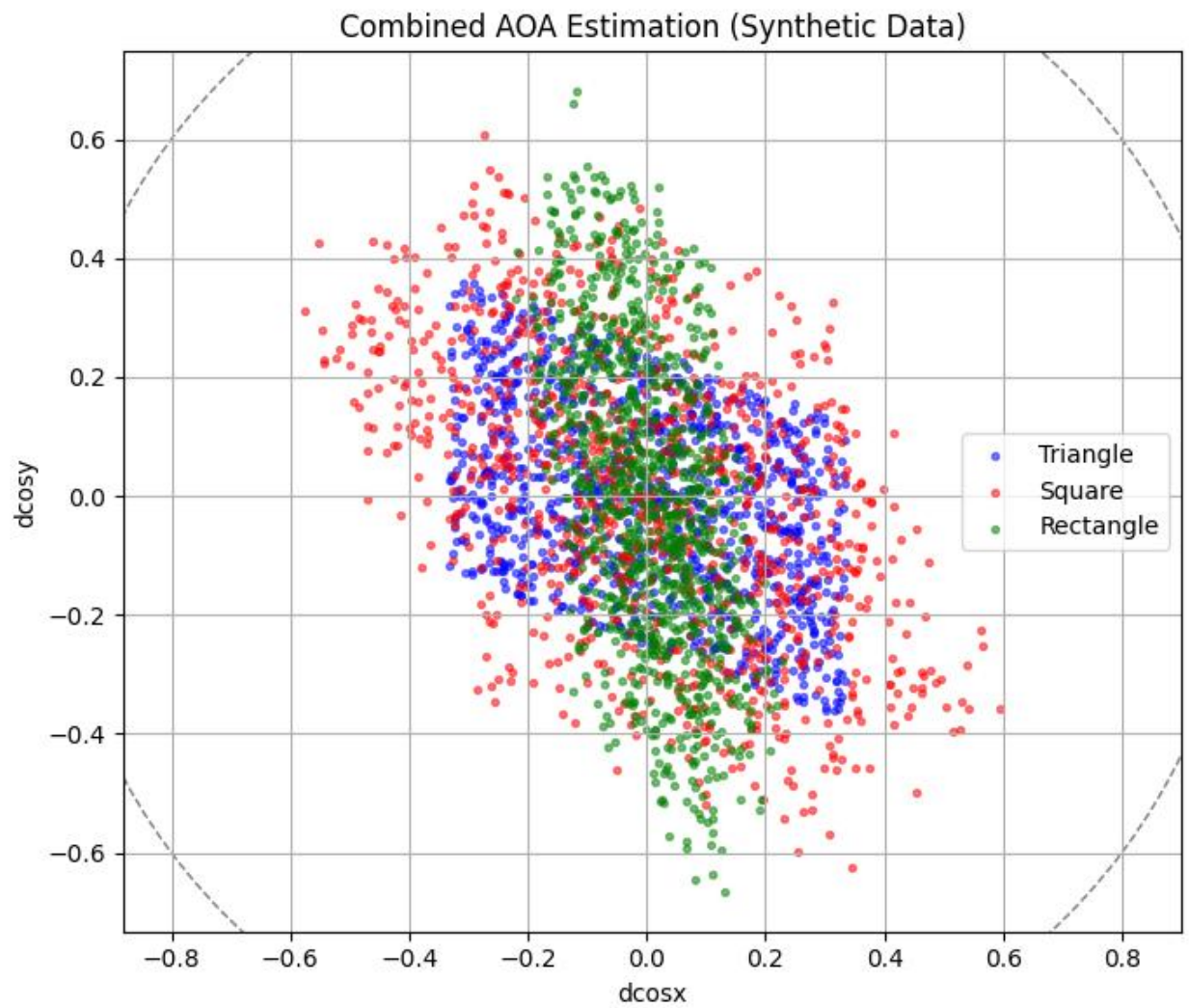
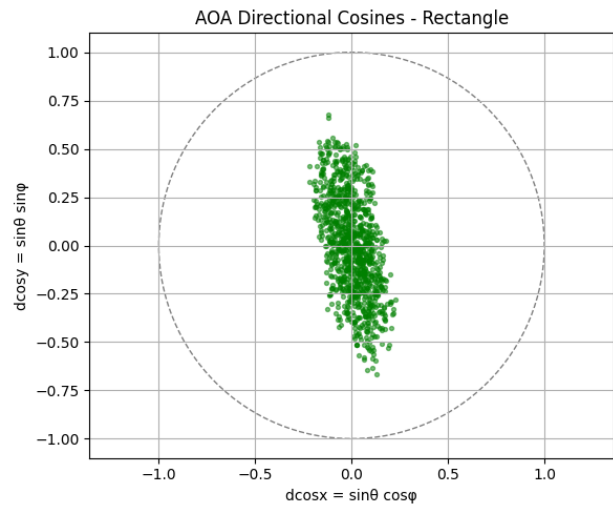
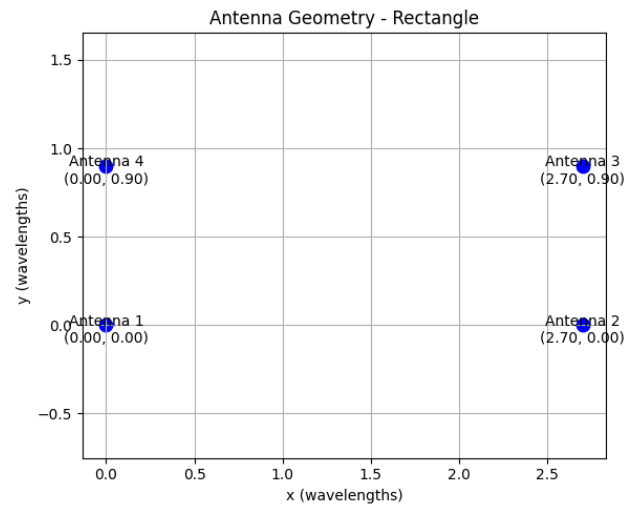
The estimated AOA points formed a tight cluster near the centre. The triangular geometry provided balanced resolution and low ambiguity.
- **Square**

The estimates covered a slightly broader area but remained well-distributed. The square layout showed symmetrical behaviour due to equal spacing in both x and y axes.
- **Rectangle**

The estimates formed a vertically stretched cluster, indicating stronger resolution in the vertical direction but more spread along y-axis due to elongated structure.

Each geometry's plot includes a circular reference to denote the maximum valid range for directional cosine values (unit circle).





Task 4e: Visible differences

In the **triangular geometry**, the estimated AOA points appear tightly clustered, indicating stable and consistent direction estimation. The symmetry of the isosceles triangle helps maintain a balanced sensitivity to both x and y directional changes, making it a good choice for general-purpose direction finding with low ambiguity.

For the **square geometry**, the AOA estimates are slightly more spread out but remain symmetrically distributed around the centre. This behaviour reflects the uniform spacing in both x and y directions, which contributes to equally good resolution along both axes. The square layout also maintains good stability and is less prone to directional bias.

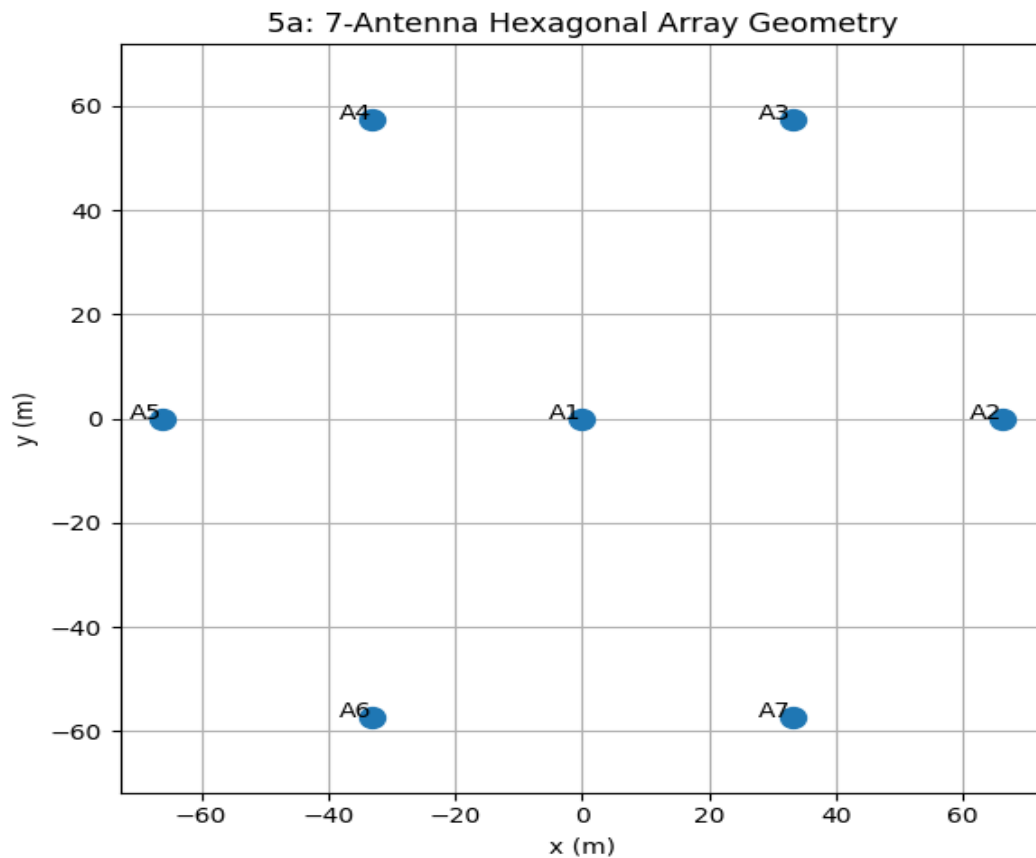
In contrast, the **rectangular geometry** produces an elongated cluster of estimated AOA points, stretched more in the vertical direction. This is due to the longer horizontal (x-axis) spacing between antennas, which enhances resolution along the x-direction but can introduce increased sensitivity to noise and phase errors in the y-direction. As a result, the rectangular array provides better angle resolution in one axis but at the cost of increased directional bias or estimation spread in the orthogonal direction.

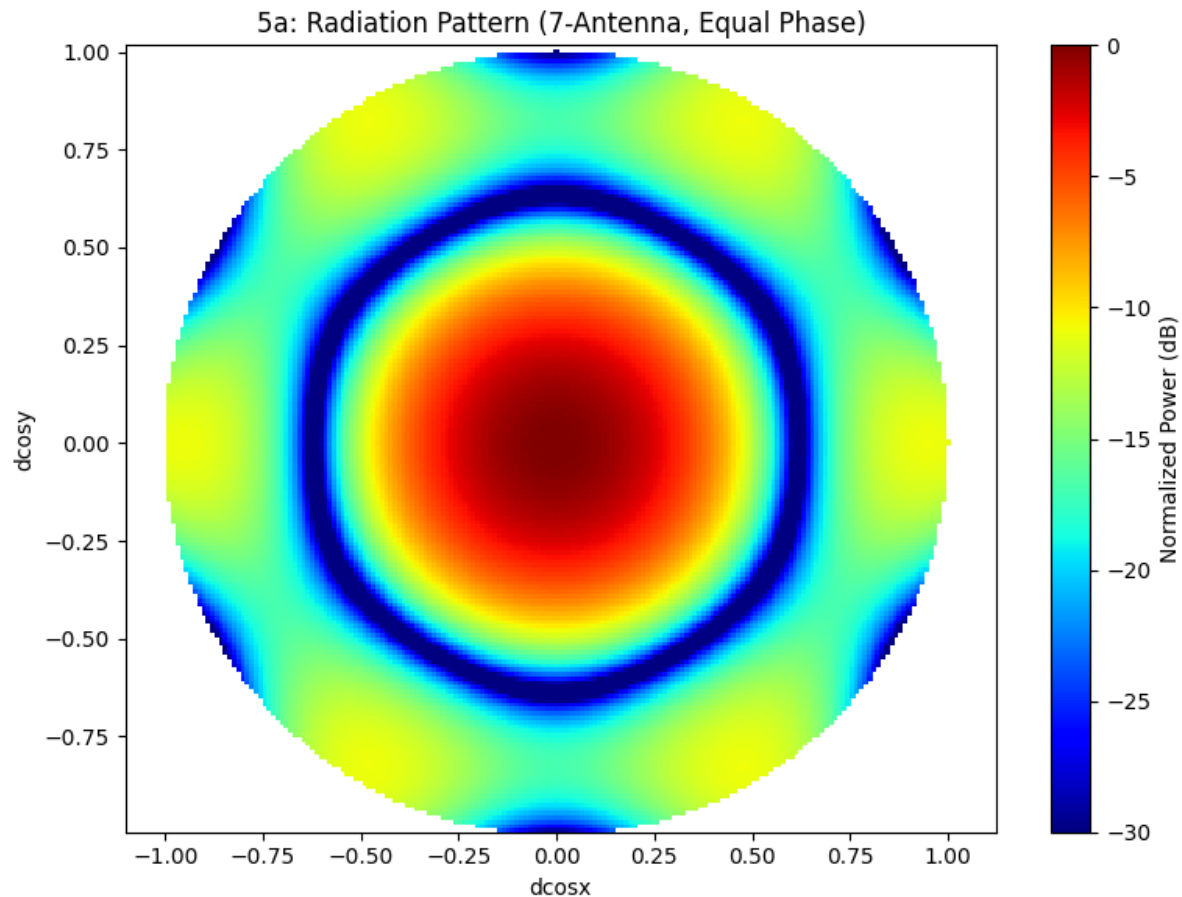
Overall, all three geometries demonstrate the ability to estimate angles of arrival using synthetic data. However, the triangle offers the most compact and reliable clustering, the square provides uniform response, and the rectangle emphasizes directional resolution but shows a larger spread due to its extended shape.

TASK5- Radiation pattern simulation of an antenna array

Task 5a: 7-Element Hexagonal Array – Equal Phase

A hexagonal array with 7 antennas was simulated. Six elements are evenly placed around a central one, each spaced 0.7λ apart. All elements were fed with equal amplitude and equal phase to observe the fundamental radiation pattern of this compact symmetric geometry.





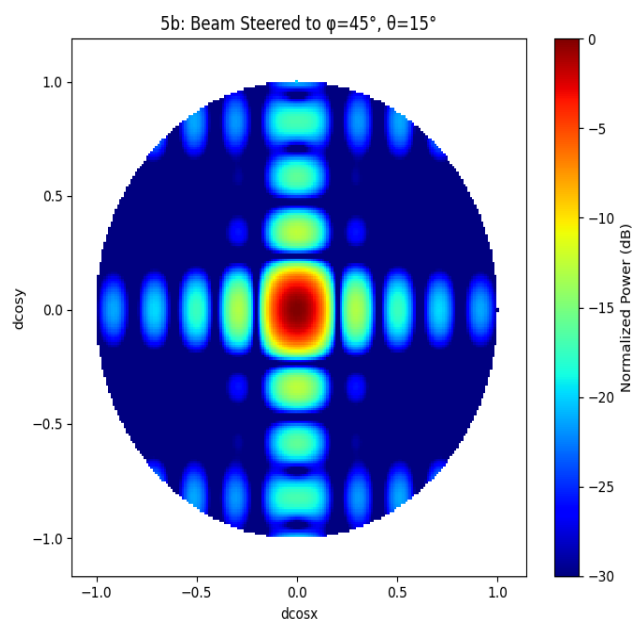
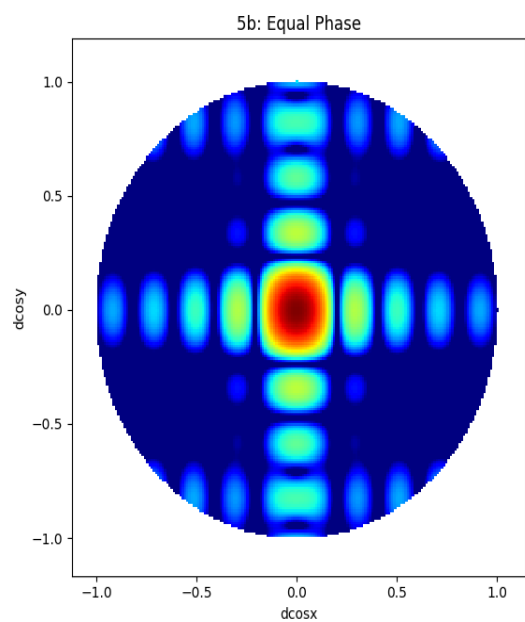
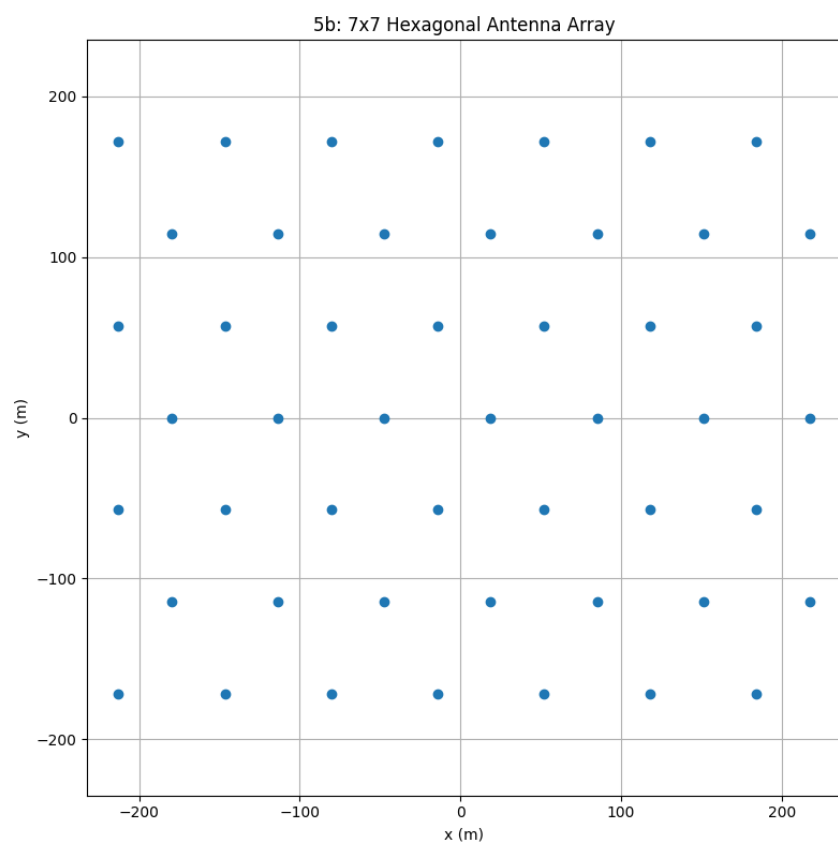
Observations

The resulting radiation pattern displayed a central dominant main lobe and symmetric side lobes surrounding it. The hexagonal geometry ensures near-omnidirectional response in the azimuth plane while preserving gain at the centre. The circular symmetry of this array offers balanced beam characteristics, but due to the small number of elements, the beamwidth remains broad. Minor side lobes exist but are relatively low, indicating acceptable directionality.

Task 5b: 7×7 Hexagonal Array – Equal Phase vs Beam Steering

To enhance directionality and control the beam, the hexagonal grid was extended to a **7×7 array** (total of 49 elements). Two cases were studied:

- **Case 1:** All elements fed with equal phase
- **Case 2:** Beam steering toward azimuth $\phi = 45^\circ$, elevation $\theta = 15^\circ$

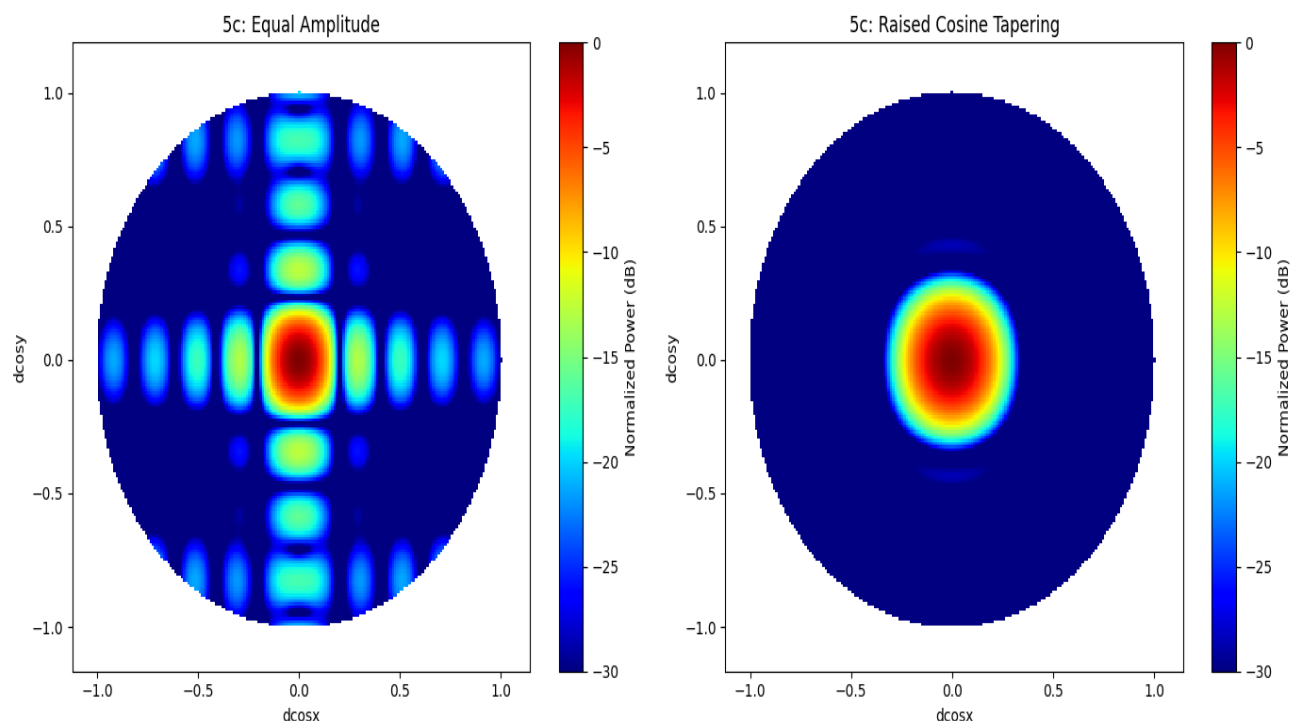


Observations

1. In the **equal phase case**, the central beam was much narrower compared to the 7-element case. This is expected due to the increased aperture size. Multiple side lobes are visible but with significantly reduced amplitude compared to the main lobe.
2. In the **beam-steered case**, the main lobe accurately shifts toward the desired direction. The simulation successfully demonstrates proper beam steering. However, side lobes become more asymmetric, and grating lobes may slightly increase due to non-zero steering angles and uniform element spacing.

Task 5c: Amplitude Tapering – Equal vs Raised Cosine

Amplitude tapering (windowing) is often used to reduce unwanted side lobes. A **raised cosine function** was applied to reduce the amplitude of outer elements, thereby minimizing energy leakage into side directions.



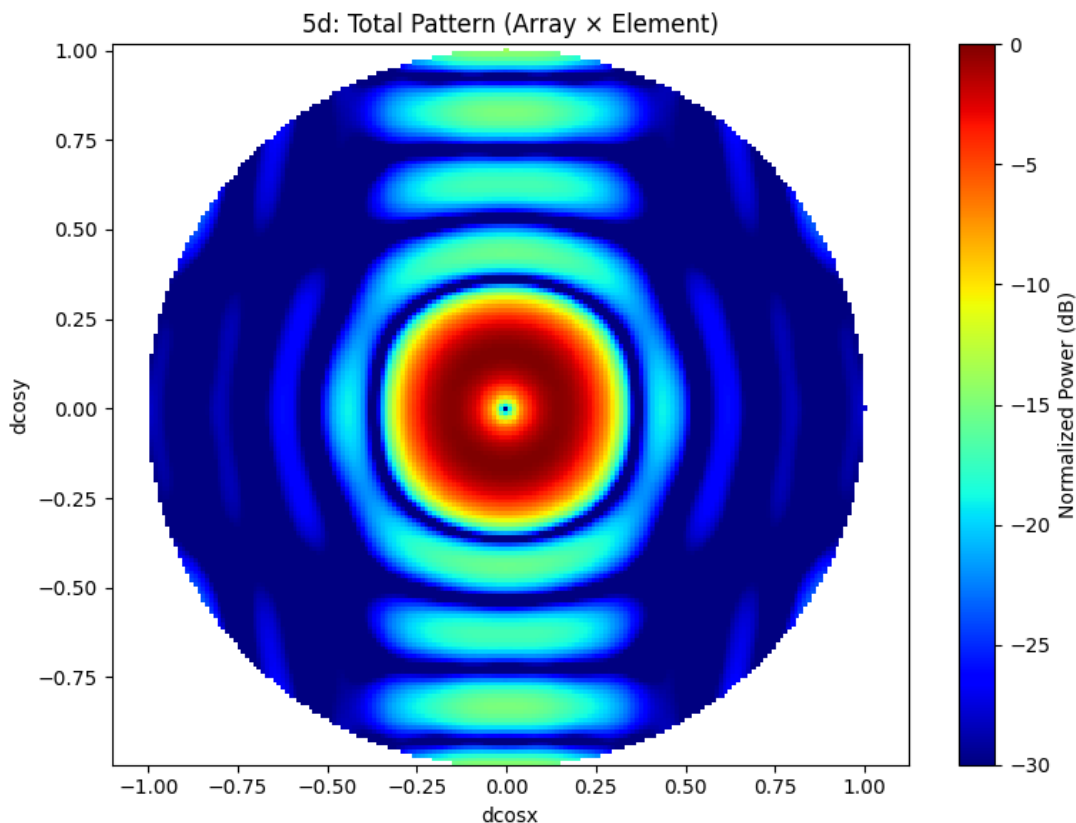
Observations

- With **equal amplitude**, the array exhibits a sharp main lobe but also shows strong side lobes, which can interfere with signal isolation.
- With **raised cosine tapering**, side lobes were greatly suppressed. The trade-off is a slightly broader main lobe, as energy is more concentrated near the centre of the array.

This clearly illustrates the classic beam-shaping trade-off: lower side lobes vs wider main beam.

Task 5d: Final Pattern = Array Factor $\times$ Dipole Element Pattern

In reality, the total radiation pattern is the **product** of the **array factor** (from antenna layout and phasing) and the **element pattern** (radiation of a single antenna). Here, a **dipole pattern ($\sin^2(\theta)$)** was used for each element to simulate a realistic case.



Observations

- The total pattern incorporates both the steering and tapering effects from previous steps, now filtered through the directional characteristics of a dipole.
- Off-axis radiation is attenuated, and the pattern appears more physically accurate. The main lobe remains dominant but its shape reflects the dipole's natural roll-off at higher θ angles.

This step reinforces the importance of considering **element radiation properties** in real-world array design, especially for narrow-beam, high-resolution radar or communication systems.

CONCLUSION

Throughout this project, we carried out an in-depth analysis of radar signal processing using both real-world and synthetic data. Starting with Task 1, we investigated raw radar returns through power plots, FFT analysis, and cross-correlation between receivers. This allowed us to identify target reflections, periodic components, and signal delays across antenna channels. Task 2 enhanced our understanding of spectral domain techniques by introducing coherent and incoherent integrations, which effectively reduced noise and highlighted stable signal features. Building on this, Task 3 focused on interferometry in the spectral domain to estimate the Angle of Arrival (AOA), where we used phase differences from cross-spectra to successfully determine spatial signal directions and visualize them in a 3D coordinate system.

In the later tasks, we moved from real data to synthetic setups for controlled simulations. Task 4 demonstrated AOA estimation using different antenna geometries—triangle, square, and rectangle—showing how element arrangement impacts directional resolution and spreads. The observations confirmed that while triangular arrays offer compact clustering, rectangular ones provide higher resolution along certain axes. Finally, in Task 5, we designed hexagonal antenna arrays and simulated their radiation patterns under various configurations—equal phase, beam steering, amplitude tapering, and total pattern formation by combining array factor with element pattern. These simulations clearly showed the effects of steering and tapering on beam shape and side lobe suppression. Altogether, the project provided a comprehensive, hands-on exploration of radar systems, antenna theory, and signal processing techniques, reinforcing key concepts through practical implementation and visualization.